

ACADEMIC COURSE MODULE

Using Evidence to Improve Health Services

From data → evidence → decision → action → learning

Professor Saloshni Naidoo

Learning objectives

- 01** Distinguish data, information, evidence and knowledge.
- 02** Match evidence to the decision a health service needs to make.
- 03** Assess evidence for validity, completeness, representativeness and uncertainty.
- 04** Recognise why good evidence does not automatically produce better services.
- 05** Use a data → evidence → decision → action → learning framework.

Opening case: something has changed

THE SIGNAL

Childhood immunisation coverage falls from 82% → 68%

Question: What should the district do?

- Increase outreach?
- Investigate stock-outs?
- Target particular facilities or communities?
- Wait until the data are validated?

The data show WHAT — not necessarily WHY or WHAT TO DO

The central idea: start with the decision



KEY INSIGHT

A good analysis is one that reduces uncertainty about a real decision.

From data to evidence

DATA

Measurements or
observations

INFORMATION

Data placed in
context

EVIDENCE

Information
relevant to a
question

DECISION

A choice among
actions

WORKED EXAMPLE

68% coverage is DATA. A decline concentrated in specific facilities becomes INFORMATION. Evidence requires asking why and what action is justified.

Evidence is decision-specific

Question

What is happening?

Why is it happening?

What might work?

Did it work here?

Useful evidence

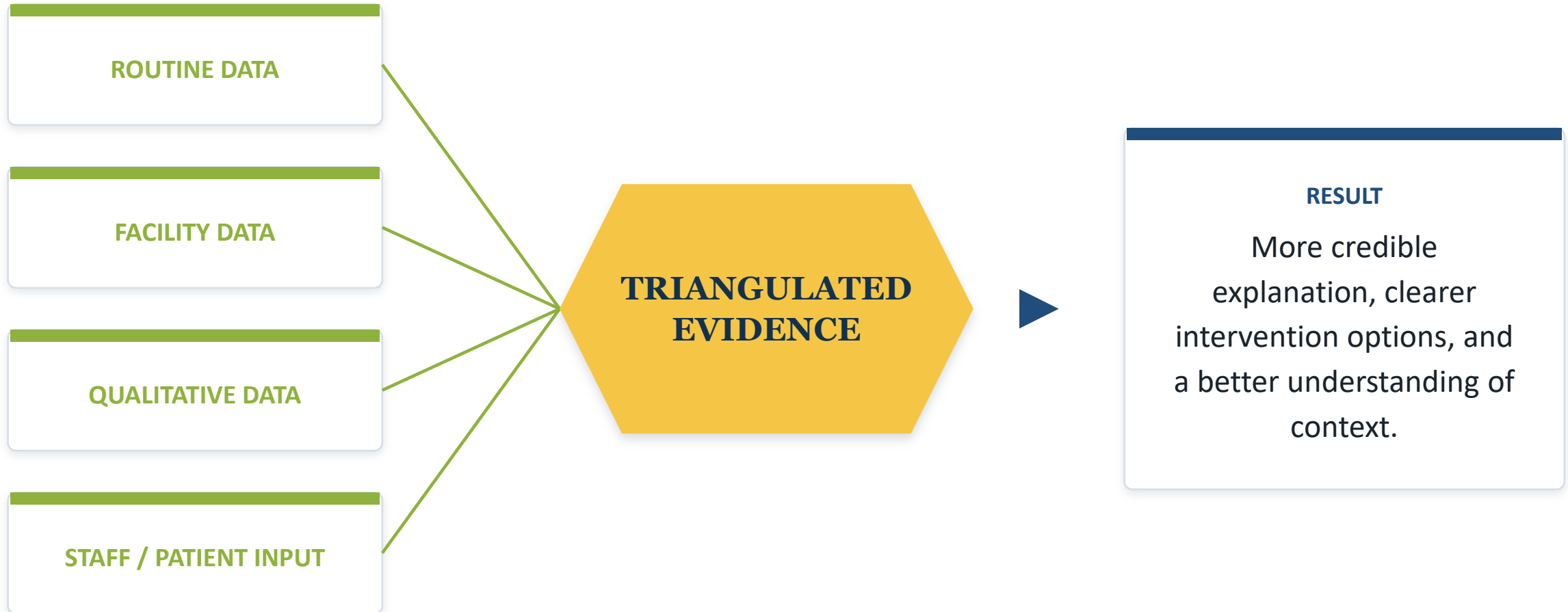
Routine data, surveys, surveillance

Epidemiology, qualitative data, process data

Trials, systematic reviews, implementation research

Evaluation and quasi-experimental designs

Triangulation: stronger evidence from multiple sources



Can we trust the evidence?

- 01 Validity** — are we measuring what we think we are measuring?
- 02 Completeness** — who or what is missing?
- 03 Representativeness** — does the dataset reflect the population?
- 04 Causality** — could other explanations account for the association?
- 05 Uncertainty** — how precise and robust is the finding?

Statistical significance $\neq$ decision relevance

The denominator can change the story

1,000 patients attended

1,000 patients attended

800 received treatment

800 received treatment

Coverage = 80%

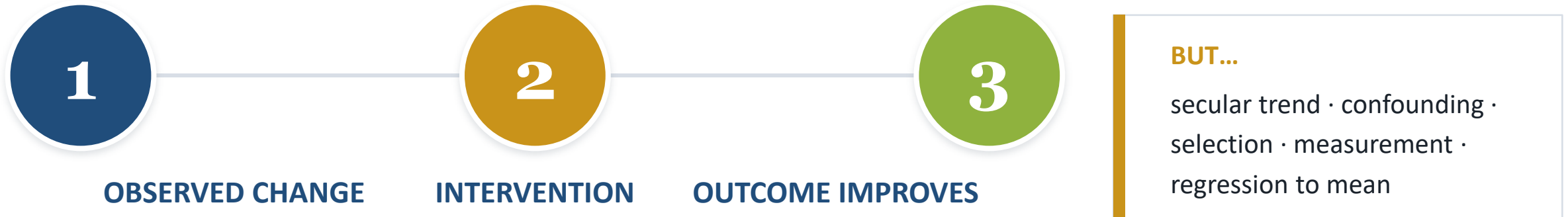
But what if 1,500 were
eligible?

$$800 / 1,500 \\ = 53\%$$

**Different denominator
→ different decision**

Always ask: what is the denominator, and who does it include?

Association is not automatically causation



—Before–after results are useful, but they often leave alternative explanations.

—Design the evaluation around the causal question.

Why evidence does not automatically improve services



Context: workforce • resources • leadership • incentives • infrastructure • acceptability • patient behaviour • governance

Can it work here, for these people, under these conditions?

Interactive case: waiting times at a district hospital

THE SIGNAL

Average outpatient waiting time: 2.1 hours → 4.3 hours

- Increase occurred mainly on Mondays and Fridays.
- Staffing levels remained relatively constant.
- Patient arrivals increased by 30%.
- Most of the increase occurs between registration and clinical assessment.

GROUP TASK · 2 MINUTES

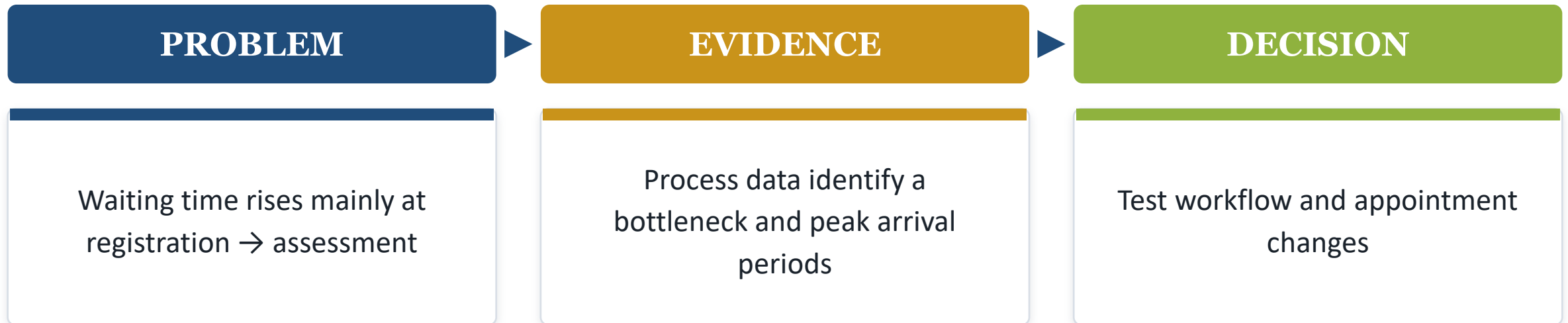
What would you recommend — and what additional data would you want?

The data scientist's next question

- Where exactly is the bottleneck?
- How do arrivals vary by hour and day?
- How long does registration take?
- How long do patients wait after registration?
- How are staffing and room availability distributed?
- Are appointment systems creating peaks?
- Are there differences by patient acuity or service type?

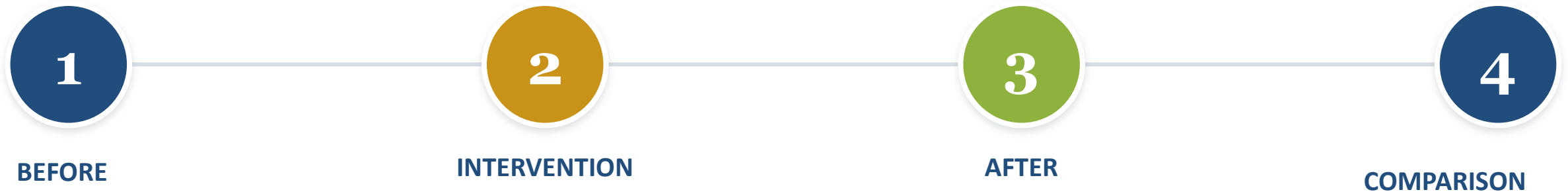
The best analysis reduces decision uncertainty.

From evidence to action: make the decision explicit



Decision ≠ recommendation from a model. It requires context, feasibility and consequences.

Did the intervention work?



- **Before–after:** simple, but alternative explanations remain.
- **Controlled before–after:** adds a comparison.
- **Interrupted time series:** examines level and trend changes.
- **Difference-in-differences:** compares changes over time between groups.

Keep measuring after implementation: evidence is a cycle, not a one-off report.

The Evidence-to-Action Cycle



Evidence improves health services when it is connected to decisions, action, evaluation and learning.

What should a data scientist do differently?

- 01 Start with the decision, not the dataset.
- 02 Make the numerator, denominator and population explicit.
- 03 Triangulate quantitative and qualitative evidence.
- 04 Distinguish description, association and causal inference.
- 05 Report uncertainty and limitations — not only point estimates.
- 06 Interpret findings in the context of the health system.
- 07 Design analysis so that results can change a decision.
- 08 Measure whether the decision actually improved the service.

Take-home message

DATA ≠ EVIDENCE
EVIDENCE ≠ DECISION
DECISION ≠ IMPLEMENTATION
IMPLEMENTATION ≠ IMPROVEMENT

The goal is a learning health service:
measure → understand → decide → act → evaluate → adapt